

12 Catalytic Reactors

Why are we learning this?

So far in this course, we have assumed that reaction rates are controlled entirely by chemical kinetics. Real catalytic reactors are more complicated. Before molecules can react on a catalytic surface, they must:

- Travel through the reactor.
- Reach the catalyst particle.
- Diffuse into the catalyst pores.
- Reach the active catalytic surface.

Each of these steps can influence the observed reaction rate. By the end of today's lesson, you should understand:

- The different length scales present in catalytic reactors.
- The difference between reaction and diffusion.
- External diffusion and internal diffusion.
- Why diffusion can limit reactor performance.
- The concept of a rate-limiting step.
- Why observed reaction rates are not always controlled by chemistry.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



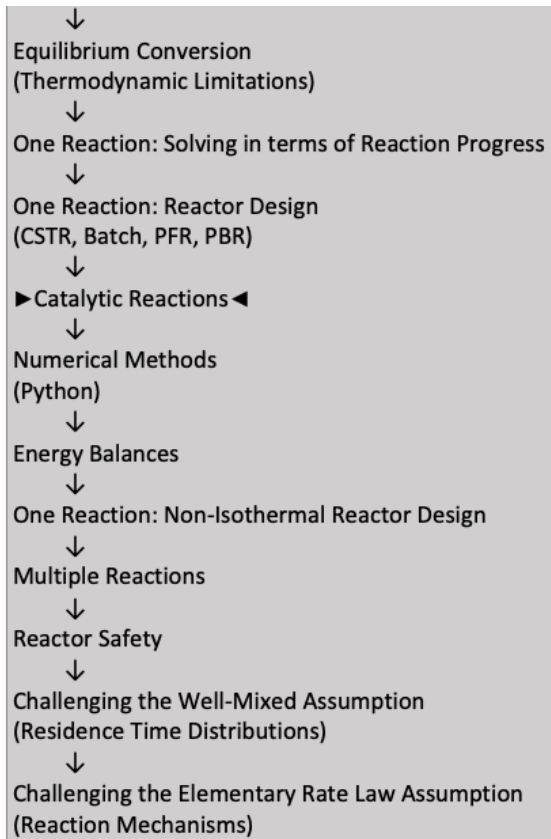
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws
(Kinetics)



Equilibrium
(Reversible Reactions)



Big Idea

A molecule must reach the catalyst surface before it can react. Transport and reaction therefore work together to determine reactor performance.

One Reactor, Many Length Scales

Catalytic reactors can be viewed at several different scales:

Reactor Level



Catalyst Pellet Level



Pore Level



Molecular Level

Different physical phenomena become important at each scale. A successful reactor design must account for behavior across all of these scales.

Why Not Just Use Powdered Catalyst?

Catalysts are often formed into pellets or particles.

Pellets:

- Provide large amounts of catalytic surface area.
- Make catalyst recovery easier.
- Reduce operational challenges such as pressure drop.

The tradeoff is that molecules must now diffuse through the pellet before reacting.

External Diffusion

External diffusion describes the transport of molecules between the bulk fluid and the catalyst surface. Typical pathway:

Bulk Fluid



Boundary Layer



Pellet Surface

Before reaction can occur, reactants must first cross this boundary layer.

Internal Diffusion

After reaching the catalyst surface, molecules may still need to travel through the catalyst pore network before reaching an active catalytic site.

Pellet Surface



Pore Network



Catalyst Particle

This process is called internal diffusion.

Reaction Is Only One Part of the Process

To convert a reactant into a product, several events must occur:

External Diffusion



Internal Diffusion



Catalytic Reaction

The slowest piece determines the overall reactor performance.

The Reaction Bottleneck Step

The bottleneck is the slowest process that occurs during the reaction. Think of a traffic jam:

Fast Road



Narrow Bridge



Fast Road

The bridge determines overall traffic flow.

Similarly:

Fast Diffusion

+

Slow Chemistry

means chemistry controls the process.

or

Fast Chemistry

+

Slow Diffusion

means diffusion controls the process.

Diffusion-Limited Reactions

A diffusion-limited reaction is one in which molecules cannot reach the catalytic surface fast enough. In this situation:

Observed Rate $\approx$ Diffusion Rate

Increasing the intrinsic reaction rate will not significantly improve reactor performance because transport is already the bottleneck.

Intrinsic Rate vs Observed Rate

Two rates can exist simultaneously:

Intrinsic Rate: The rate determined by chemistry at the active site.

Observed Rate: The rate actually measured in the reactor.

When diffusion limitations exist:

Observed Rate $<$ Intrinsic Rate

because transport prevents reactants from reaching catalytic sites rapidly enough.

Summary and Key Takeaways

Today we introduced catalytic reactors and the transport processes that occur within them.

Key Ideas

- ✓ Catalytic reactors involve several length scales, from the reactor level down to the molecular level.
- ✓ Molecules must reach catalytic surfaces before reaction can occur.
- ✓ External diffusion describes transport through the boundary layer surrounding a catalyst particle.
- ✓ Internal diffusion describes transport through the catalyst pore network.
- ✓ Reaction and diffusion can both influence the observed rate.
- ✓ The slowest process acts as the reaction bottleneck and controls overall performance.
- ✓ Diffusion-limited systems are controlled by transport rather than chemistry.

Most Important Idea

Observed Rate = Reaction + Transport

A catalytic reactor can only perform as fast as its slowest step. Understanding both diffusion and reaction is therefore essential for catalytic reactor design.